

Cor fn convexa martingala submartingala

Corolario 1. Sea $(X_n)_{n \in \mathbb{N}}$ una martingala y $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ convexa

$\implies (\varphi(X_n))_{n \in \mathbb{N}}$ es una submartingala.

Demostración: Por la desigualdad de Jensen condicionada, tenemos que

$$\varphi(X_n) = \varphi(\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]) \leq \mathbb{E}[\varphi(X_{n+1}) \mid \mathcal{F}_n] \text{ c.s.}$$

$\uparrow$
 $(X_n)_n$ martingala

Por tanto, $\varphi(X)$ es una submartingala respecto a la filtración $\mathcal{F} = (\mathcal{F}_n)_{n \in \mathbb{N}}$. ■

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